

**CENTRE FOR DEVELOPMENT OF IMAGING TECHNOLOGY
TECHNOLOGY EXTENSION DIVISION**

SYLLABUS

DIPLOMA IN COMPUTER APPLICATION (DCA)	
Scheme: 2023 Scheme Duration: 6 Months Eligibility: SSLC or Equivalent	
THEORY SUBJECTS	
DCA 101	Introduction to Information System
DCA 102	Networking and Internet
DCA 103	Office Automation Systems
DCA 104	Principles of Programming and Web Technologies
LAB SUBJECTS	
DCA 105	Lab I-Computer Basics and Office Automation
DCA 106	Lab II- Programming in C and Python & Web Application Development
ANNEXURE	
Assignments for Theory & Questions for Lab	

Detailed Syllabus(Theory)

DCA 101 - INTRODUCTION TO INFORMATION SYSTEM

Theory Hours - 50 Hrs

Module I - Computer Fundamentals (10Hrs)

Learning outcome

- Understanding of the concept and architecture of computers.
- How a computer system works?
- Learning the basic terminologies.

Topics

Introduction :- Fundamentals - Definition, Hardware, Software, Operating System, data, information etc; Functions and Characteristics; Applications of Computers; Capabilities and Limitations

Familiarise different generations of computer

Computer classifications :- Analogue / Digital / Hybrid, PC, Microcomputer, Minicomputer, Mainframe computer, Super Computer.

Components of a computer :- Input Unit ; Output Unit ; CPU ; Memory unit

Input devices :- Keyboard; Mouse; Scanner; Digital/ Web Camera; Touch screen; graphic tablet

Output Devices :- VDU; Printer; Audio Output devices; LCD Projector

Intelligent terminal and dumb terminal

Memory Unit ; Volatile and Non-Volatile memory ; Primary Memory ; Secondary Memory ; BIOS Memory ; Cache Memory ; Virtual memory ; SIMM ; DIMM ; Storage devices; online / offline storage.

Familiarisation of other hardware components :- SMPS ; Motherboard ; Video Cards - PCI, PCI Express, and AGP ; Multimedia Cards - Sound Cards, TV Tuner Cards and Capture Cards ; I/O Cards - SCSI, Serial, USB, and Parallel ; Communications Cards - NICs and Modems

Ports :- Characteristics and different types - Serial, Parallel, USB, Ethernet port etc

Configuring the Peripheral Devices ; methods for connecting peripherals to system - different types of cable connection, explain Bluetooth and Wi-Fi / IR, NFC connectivity

Data representation in computers :- Number System - Types, examples, Conversion rules ; Coding system - ASCII, BCD, EBCDIC, ISCII, Parity Bit, etc.

Units :- Time Units - Milliseconds, Microseconds, Nanoseconds, Pico seconds ; storage units - Bit, Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte, Exa Byte, Nibble.

Module II - Basics of Information Technology (10Hrs)

Learning outcome

- Acknowledge the role of technologies in today's society
- To identify the basic hardware and software components of a computer and explore their functioning
- Learn about various operating systems

Topics

Introduction to IT - Role of IT in society - Advantages and Disadvantages, Characteristics and features of IT, Application areas - Science and Engineering, Education, Business and Commerce, Medicine and Entertainment – Advantages & Disadvantages of IT

Basics of Computer System - Generations & Peripheral devices - Hardware - Software - system software & application software-off the shelf and custom software-programming languages-compilers-interpreters- Operating systems: structure & functions - Types of Operating system: Batch processing OS, Mutitasking OS, Multiprocessing OS, Time sharing OS, Real Time OS, Distributed OS, Network OS-Examples of OS: MS-DOS, MS-Windows, Linux Ubuntu, Apple Macintosh

Data and information Concepts -- Importance of data and information, Difference between data and information

Introduction to ITES (IT Enabled Services) - what are IT enabled services? Benefits and threats, ITES fields - BPO - GIS - Call Centers - Data Management Services - Medical Transcription - Data Digitization, Web services, Payroll management, Telemarketing

Module III - Basics of Software (10Hrs)

Learning outcome

- Distinguish System software and Application software.
- Commonly used operating systems.
- Identify the primary functions of Operating Systems.
- Installation and configuration of software.

Topics

Types of Software :- Software - System software , Application software, Utility Software ; Demo ware ; Shareware ; Freeware ; Firmware ; Free and Open Source Software.

Operating Systems :- Functions, Types - Batch Processing , Single User, Multi User, Multiprogramming, Multi-Tasking (Multiprocessing); Familiarise Multithreading

Booting and Shutdown of Computers ; BIOS ; POST ; Master Boot Record(MBR)

Introduction to old versions of Windows OS

Detailed study of Windows :- Overview ; GUI basics (Desktop, taskbar, icons, desktop background management) ; Navigation (Start menu, File explorer) ; Accessories ; Control Panel ; User Account management ; storage management ; file and folder management ; keyboard shortcuts ; data backup

Familiarise Ubuntu Linux :- Introduction ; History ; Versions ; basics

Familiarising Command mode :- Windows (DOS) ; Linux (Terminal)

Familiarize other operating systems :- UNIX ; Apple Macintosh

Linux commands

Computer Virus :- Introduction ; Mechanism of Virus ; How a virus spreads ; How is virus named ; A few prominent viruses ; Types of Computer Virus ; Related Concepts - Antivirus programs - Norton antivirus program

Module IV - Basics of Information System (10Hrs)

Learning outcome

- Knowledge about information system, its capabilities and uses with IT infrastructure in a business enterprise to achieve desired goals.

Topics

Information System :- Data, Information & System ; Information System; Dimensions of Information System - Organization, Management, IT ; Functions of Information System

Business Information Systems :- Role of Business Information Systems - Top, Middle, Bottom ;

Types of Business Information Systems - TPS, MIS, EIS, DSS, AI/Expert Systems

Management Information System(MIS) :- Objectives ; Components ; Functions ; Resources ; Characteristics; Terminologies and Applications

Enterprise System and Types :- ERP ; CRM ; SCM

Module V – Basics of Databases (10Hrs)

Learning outcome

- Knowledge about data processing, data management systems, relational data model, familiarize MS Access, SQL and importing of Excel files.

Topics

Database Management System – Introduction; Data Processing Vs. Data Management Systems ; Database Concepts; Characteristics of Database; Types of Database System; Advantages and Disadvantages of a DBMS ; Database Administrators and Database Users

Creating database using MS Access - Importing Excel files to MS Access; Using Forms-creating a form and editing data using form

Relational Model Concepts - Data Definition Language (DDL) ; Data Manipulation Language (DML); Introduction to SQL; SQL commands; Importing Excel files to SQL; Familiarise MySQL, NoSQL and Oracle

DCA 102 - NETWORKING AND INTERNET

Theory Hours - 50 Hrs

Module I - Computer Networks (10Hrs)

Learning outcome

- Knowledge about networking technologies and types of networks, devices used, security devices, topologies, nodes, mobile communication and latest technologies

Topics

Introduction to Computer Networks - Evolution of Networking, Wired & Wireless Networking

Types of Networks - LAN (Local Area Network) , MAN (Metropolitan Area Network) , WAN (Wide Area Network) , PAN (Personal Area Network)

Network Devices - Hub, Switch, Router, Bridge, Gateway; Modem, Repeater, Access Point; Modem, Ethernet Card, RJ11, RJ45

Network Security Devices - Active Devices, Passive Devices, Preventative Devices, Unified Threat Management (UTM), Firewalls

Integrated Services Digital Network (ISDN)

Networking Topologies - Bus Topology, Star Topology, Ring Topology, Mesh Topology, Tree or Hybrid Topology

Nodes in a Networked Communication - Types of Nodes, MAC Address, IP Address, IPv4, IPv6

Connecting to Internet - Connecting the computer to the Internet

Types of connectivity - Dial-up; Wired broadband connectivity - ISDN, Cable Internet, DSL, Leased Line, FTTH; Wireless broadband connectivity - Mobile broadband, Wi-MAX, Satellite broadband

Internet Access sharing methods - LAN, Wi-Fi, Li-Fi

Services on Internet - World Wide Web (WWW), Browser, Web browsing, Search Engines, E-mail, Social media; Various stages of Data Transfer from Server to Client; Domain Name; Resolution; Mobile communication;

Generations in mobile communication - First Generation, Second Generation - GSM, GPRS, EDGE, CDMA, Third Generation - WCDMA, Fourth Generation, Fifth Generation; Mobile communication services- SMS, MMS, GPS, Smart Cards

Mobile Operating System

Latest Technologies - Data Analysis; Cloud Computing; Internet of Things & Edge Computing; Block Chain; Artificial Intelligence & Machine Learning; Augmented Reality & Virtual Reality; Robotic Process Automation (RPA); 5G Spectrum

Module II - Introduction to Internet (10Hrs)

Learning outcome

- Understanding about internet
- Explore different kinds of information available on the Internet, surfing safely and Searching information on the Internet
- Importance of online conference tools in daily life and its usage
- Understand e-learning concept of e-learning and its importance in the modern world.

Topics

Concepts of Internet – Basics of internet connectivity -Applications of Internet –tools and troubleshooting– Intranet and extranet -Internet Services-communication services- Information retrieval services-Web services-World Wide Web-email– Web Browsing softwares – Search Engines – URL – Web Protocols- Domain name – IP Address – Surfing the web

Basics of electronic mail –concept, technology behind email, different email providers, Getting an email account – Sending and receiving emails – mail operations –email folders-applications of Emails-web mail-client mail-email security- email client(eg-outlook)

Online/Web/Mobile Conferences & Tools – introduction towards Audio and Video conferencing – Advantages & Disadvantages -Popular Video conferencing tools

eLearning – What is eLearning, Features and uses, Different eLearning platforms in India, LMS

Learning outcome

- Understand the concept of e-government and the associated benefits and drawbacks
- Gather knowledge on various National and State e-governance initiatives.
- Recognize the benefits and limitations of e-commerce
- Understand major e-commerce business models and their impact in the society

Topics

What is E-Governance, objective, Scope, Benefits and outcomes of e-Governance - **Pillars of e-Governance:** People - Process - Technology - Resources, **Types of interactions in e-Governance:** Government to Citizen (G2C)- Government to Businesses (G2B)-Government to Government (G2G)-Government to Employees (G2E), **Phases of e-Governance Maturity Model** (Phase I: Information, Phase II: Interaction, Phase III- Transaction - Phase IV : Transformation)

e-Governance Infrastructures: Common Service Centre (CSC) - State Wide Area Network (SWAN) - National e-Governance Service Delivery Gateway (NSDG) - State Data Center (SDC)

Challenges to e-Governance, Security approaches to e-Governance

National level e-Governance initiative - Digital India initiative, Major services and projects implemented by union govt and state govt.

e-Commerce (Internet Commerce or electronic commerce) - What is e-Commerce - Features and Scope of e-Commerce - Traditional Commerce vs e-Commerce - e-Commerce Models (B2C, B2B, C2C, C2B, B2G - Benefits and limitations of e-Commerce - Major e-Commerce websites and its usage

Module IV- Mobile Computing and Digital Payment System (10Hrs)

Learning outcome

- Understand mobile computing, its principles and theories.
- List applications in different domains that mobile computing offers to the public, employees, and businesses.
- Understand the various options of payments - online & mobile payment, smart cards etc

Topics

Mobile Computing - Introduction and Concepts- mobile communication, mobile hardware and software, Mobile Device Operating Systems – Android, iOS, - Generations of Mobile Communication Technologies

e-Commerce & m-Commerce System Security

Familiarisation of mobile applications - Introduction to mobile applications-types of mobile applications-native web and hybrid, Different categories of mobile applications - cooking application - Education applications - Communication applications - Shopping application etc

Digital economy - Identify the methods of payments on the internet – e-Cash , e-cheques , credit cards on the Internet

Electronic Payment Systems - Need and uses of Electronic Payment System- Electronic Fund Transfer-one time password (OTP)-PIN and its importance-internet banking – Credit card - Debit card - UPI, Point of Sale(POS), Security Issues

Mobile Payment System - Different payment types: Mobile Wallets, QR code payments, Payment links, SMS payments, Benefits of Mobile Payment

Security issues in digital payment system ,Online banking frauds

Do's and Don'ts in electronic/mobile payment system

Learning outcome

- Understand various social media technologies and their importance in the day to day life.
- Benefits of social media in digital marketing and Education,
- Awareness on various cybercrimes and cyber security,
- Understand copyright and intellectual property rights

Topics

Social Media – Why & what Social Media – Present and Past - Social media Platforms and applications – Usage in business & Education- social media etiquette

Social media and digital marketing

Information Security – Concepts of Security - two factor authentication(2FA) - Biometric authentication methods - Data Encryption and Decryption- Cryptography

Awareness about cyber crime and Cyber security

Cyber crime - What is cyber crime, Understand the term computer Virus and Malware, Email spamming, Email spoofing, Email bombing, Forgery, Hacking, Cyber Stalking or Bullying, Web jacking, Phishing, Pornography, Cookies and threats

Cyber security - Basic concepts - Elements of security - Data security and protection from cyber attack - Antivirus - password, Concept of Firewall, Digital signature - uses and working

Cyber Laws - An Overview & Scope of Cyber Law - Copyright, Trademarks and Software Patenting - Data Privacy & Confidentiality - Cyber laws in India - IT ACT-objectives, features, offences

DCA 103 - OFFICE AUTOMATION SYSTEMS

Theory Hours - 50 Hrs

Module I - Word Processing and Indian Language Computing (10Hrs)

Learning outcome

- Work with MS Word and Open Office Writer.
- Understand the features affecting the page layout.
- Understand to use Indian languages in preparation of documents.

Topics

Introduction to Word Processing - Word Processor Basics, typing text, selecting, copying and deleting text

Creating and formatting documents using MS - Word :- MS Word Interface, Toolbars, Ribbon and Menus; Creating a new word document; Clipboard, Cut, Copy, Paste text; Find, Replace and select Text; Insert and delete comment; Use of Undo and Redo operations; Saving a document; Opening and closing a document

Text Formatting - Familiarise various Fonts, font size, font style; Format text, Bold, Italic, Underline, Strikethrough, Superscript and subscript; Text effects, text highlighting, applying font color; Change the Text Case; Alignment of text; Formatting a paragraph, Drop Cap, spacing, Tabs and Indents ; Copy Formatting with format painter; Using Borders and Shading; Create a numbered list or bulleted list

Working with objects - Insert pictures, shapes, symbols, Date and Time; Insert Text box, charts, Word Art, symbols; Insert Header and Footer, Footnote and End note; Page numbering; Insert a manual page break

Document Formatting - Formatting using themes; Using watermark; Applying Page Borders

Page Layout Settings - Page setup options – Setting Margins, Page orientation, Paper size; Creating documents with columns

Document Review - Checking spelling and grammar in a document; Using of Thesaurus; Auto correct; Tracking changes

Viewing a document - Different views of a document; Zooming; Arranging windows; Creating macros

Styles - Apply a style; Apply a document theme; View all available styles; Change style set; Create a style

Navigation - Open the Navigation pane; Search for a word or phrase; Search for Graphics, Tables, Equations or Comments; View search results; View documents headings; View a document's pages; Adding Hyperlink; Inserting Bookmark

Drawing and Graphics - Insert a picture, screen shot, shapes; Adjust text wrapping; Format, Resize and delete an object

Working with Tables - Insert a table; Manipulating Rows & Columns; Merging & splitting cells ; Adjust Column width or row height; Text Direction

Mail Merging - Creating a Mail Merge Document; Creating address list; Merging Main Document with data, Saving

Printing a document - Previewing, settings - Print All / Current Page / Custom Print; Printing a document; Advanced Printing Options

Converting documents to other file formats(eg- pdf)

MS Word Keyboard shortcuts

Word Processing using Open Office Writer - Working with documents; Formatting documents; Working with Tables; Inserting Pictures/Files etc; Tools - Spell check, macros, mail merge etc; AutoCorrect; Manually running the Spell checker; Finding Synonyms with the Thesaurus; Finding items in a Document; Create and modify Headers and Footers; Create and modify Page numbers; Adding Graphics, Formatting a Picture; Resize a picture, Wrapping a picture around text; Adding borders and colors; Creating a Form Letters; Creating a Data Source; Reviewing the Mail Merge document; Addressing Mailing Labels

Language Computing - Complexities of Indian languages; Pre -Unicode Era; Advantages of Font Encoding; Disadvantages of Font Encoding; What is Unicode? History, Format, varying length encoding; Packages and tools available under Windows for Indian languages; Difference between Unicode and non-Unicode Fonts; Familiarisation of Malayalam typing keyboard

Learning outcome

- Examine spreadsheet concepts and explore the Microsoft Office Excel environment, import and export data

Topics

Introduction to Spreadsheets - Spreadsheet Basics; Use of spreadsheets

MS-EXCEL- Basics; Familiarise MS- Excel Interface; Components of an Excel Workbook; Worksheet, Cell and Cell Address; Create New Workbook; Typing data; Saving and opening and closing a workbook; Preview and print workbook; Undo and Redo operations in workbook; Selecting a cell, cell range, entire worksheet; Different views; Zooming

Spreadsheet manipulations - Editing cell contents; Cut or Copy Data ; Previewing an Item Before Pasting; Paste Special options; Move or Copy Cells Using Drag and Drop; Copy Using Auto Fill; Complete a Series Using AutoFill; Insert a Column or Row; Delete a Column or Row; Insert a Comment; Copying and moving Spreadsheet; Naming, inserting and deleting Spreadsheet

Formatting - Formatting Text – setting Font, font size, style; Setting font color, Bold, Italic, Underline cell contents; Text Wrap, Merge & Center; Format Cell values; Copy Formatting with the Format Painter; Cell Alignments; Applying borders; Apply a Cell Style; Format a Cell Range as a Table; Apply a Document Theme; Conditional Formatting ; Page setup options; Setting header and footer in a worksheet; Inserting page number and page breaks in a worksheet; Setting background in a worksheet; Applying Borders and shading; Fill Series; Sorting & Filtering of data ; Using Auto Sum

Calculations in a spreadsheet - Introduction to formulas and Functions; Creating a formula; Using Functions –various functions in MS Excel and its use; Use of Vlookup function; Reference a Cell in a Formula; Create an Absolute Cell Reference

Sorting and Filtering data in a spreadsheet - Data Sorting; Data Filtering and Advanced filtering; Data Validation; Introduction to Pivot Table and create a Pivot table

Working with Charts - Introduction to Chart and its types; Inserting Column and Pie chart; Manipulating charts

Mail merging using worksheet data - Importing address list from Excel file for mail merging

Workbook Views - Various views of a workbook; Freezing Panes; Using macros in a workbook

Protecting the workbook - Protecting a workbook with a password

Printing the worksheet - Page Setup; Setting Print Area and Print Titles; Print Preview and Print options

MS Excel Keyboard shortcuts

Module III - Presentations (10 Hrs)

Learning outcome

- Knowledge about PowerPoint interface.
- Creating multimedia presentations.
- Using design layouts and templates for presentations.

Topics

Create and Manage Presentations using MS-POWERPOINT - The Basics; Introduction to Presentations and slides; Familiarizing PowerPoint Window; Create a New Presentation; Creating a presentation using templates; Open a Presentation; Saving a Presentation

Working with Slides - Inserting a new slide; Selecting a slide layout; Copying slides; Setting slide background; Applying themes to slide layout; Making duplicate slides

Manipulation of Slide contents - Inserting Text Boxes; Formatting text; Creating list with bullets and numbering; Changing Text directions; Inserting pictures, Shapes, Word Art; Inserting Chart inside a slide; Grouping objects in a slide; Inserting slide header and Footer; Different views

Transitions and Animations - Applying various Slide; Transitions; Animate slide contents, custom animation; Set timing for Transitions and Animations; Creating multimedia presentation - add video and audio

Slide Show - Running a slide show; Creating custom slide show

Sharing A Presentation - Convert presentation to PDF ; Convert presentation to Video ; Create an Image from presentation ; Print a PowerPoint Presentation; Broadcast PowerPoint presentation

Open Office Impress - Impress characteristics; Start Impress, Presentation Wizard; Drawing, Zoom, Lines, Rectangle, Freeform line filled, Polygon, Convert objects, Curves; Select objects, Copy objects, Delete objects, Flip objects, Arrange objects, Group objects; Export drawing, Area Fills, Color, Gradient, Hatching, Bitmaps, Transfer settings, Images, Insert images, Filters, Graphics mode, Color, Transparency, Crop, Gallery, Glue points, Size and position, Rotate, Alignment, Text, Text to polygon, Text boxes, Slides, Insert, Clone, Rename, Delete slides; Format, Character, Paragraph, Textbox format, Slide effects, Slide transitions, Sounds ; Automatic transition, Manual transition, Hyperlinks, From text, From images, Bitmaps

Module IV- Basics of Accounting & Billing Systems (10Hrs)

Learning outcome

- After going this module one should get an idea about basic accounting components and can familiarize billing system

Topics

Basics of Accounting – Accounting; Business Transaction; Purchase; Sales; Purchase return (Outward invoice); Sales return (Inward invoice); Assets; Liabilities; Capital; Debtors; Creditor; Debit; Credit; Drawings; Receipts; Account; Ledger; Journal entries; Trial balance; Profit; Profit & loss account; Balance Sheet

Billing System – Introduction; Need of Billing System; Features of Billing System; GST Billing; Billing using MS-Excel

Module V - Printing Technologies & Multimedia (10Hrs)

Learning outcome

- Knowledge about modern printing technologies, DTP, familiarize with image tools and Multimedia techniques

Topics

Printing Technologies - Introduction to various modern Printing processes; Printing Technology using DTP; Advantages of DTP; Introduction to DTP software; Use of various

tools in MS Paint, GIMP; Importing Graphics; Image Compression; Electronic Image file formats- BMP, JPEG, TIFF, GIF & PNG ; Topographic design; Vector drawing techniques; Portable Document Format (PDF)

Multimedia - Definition; Multimedia systems, elements & applications; Multimedia system architecture; Digital media and Hyper media; Multimedia file formats & standards; Types and methods of compression and decompression; Multimedia I/O Technologies; Adding text, integrating audio with Video; Image enhancing designing techniques; Image authoring and editing tools

DCA 104 - PRINCIPLES OF PROGRAMMING AND WEB TECHNOLOGIES

Theory Hours - 50 Hrs

Module I - Basics of Programming (10Hrs)

Learning outcome

- Understand the methodology to solve the problem, algorithm, flowchart & coding
- To identify programming language classifications.
- To Learn Object oriented programming concepts.
- Understand the features & architecture of Microprocessor generations and types.

Topics

Algorithms - Properties of a good algorithm - Flowchart - Problem-solving using Computer - Introduction to Programming Languages - Machine Language - Assembly Language - High Level Programming Languages - Procedure Orientation and Object Orientation - Object Orientation concepts - Compiler and interpreter - Introduction to Microprocessor & Microcontroller Programming Operation Cycle (Fetch, Decode, Execute, Store)

Module II- Introduction to C Programming (10Hrs)

Learning outcome

- To provide an overview of working principles of C language.
- Construct, compile, link & execute the C programs
- To learn and implement C language programming techniques.

Topics

Introductions to C programming - Structure of a C program - Input and output Statements - Keywords-Data types - Constants - Variables - Declaring Variable Names - Operators and Expressions - Operator Precedence in Expressions - Increment and Decrement Operators - Conditional Statements - simple if - if-else - switch - Implementing Loops in Programs - The While Loop - The do while Loop - The for Loop - Break and Continue Statement

Module III - C Programming: Arrays and Functions (10Hrs)

Learning outcome

- To understand and apply the functions and arrays.
- To implement the features of C language in real world applications

Topics

Arrays - Defining and Manipulating Arrays - Array Variable - Syntax Rules for Arrays - Reading and Writing Multidimensional Arrays.
Functions - Functions Definition- Function Calling - Recursive functions - Character Strings -The Character Data Type - Manipulating Strings of Characters- Arrays in Functions.

Module IV - Web Programming: HTML, CSS and JavaScript (10Hrs)

Learning outcome

- To understand the working of a web application.
- Understanding the use of HTML, CSS and JavaScript in websites
- To develop websites.

Topics

Introduction to web technology - Web application- Web server and Application server - Client and Server - Scripting languages

HTML and HTML5 - HTML Editors - Elements - Attributes - Headings - Paragraphs - Formatting - HTML Comments - Colors - Fonts - Hyperlink- Images - Tables - Lists -- Iframes - Marquee - Forms - Embedding Audio and video

CSS3 Introduction – Syntax - Selectors - External, Internal & Inline Style sheets - Backgrounds - Colors - Text - Fonts - Links - Border - Outline – Margin - Padding - Align - Positioning – Gradients

JavaScript introduction - Syntax - External & Internal usage - Variables - Operators -- Functions - Events - Comparison - Condition -- Loops - Dialog box - Form Validation.

Module V - Introduction to PHP and Python Programming (10Hrs)

Learning outcome

- Understand the features and uses of PHP
- To be able to create web pages using PHP.
- To get a basic understanding on web servers.
- To be able to write programs in the Python programming language.

Topics

Introduction to PHP - Advantages of PHP - Where to use PHP - Installing PHP - PHP Syntax - Variables - Echo and Print - Data Types – Constant - Operators - IF Statement - Switch statement - Loops.

Introduction to Python - Data Types - Variables - Basic Input-Output Operations- Keywords- Identifiers - Literals - Punctuations - Operators - Basic loops and conditions - String - List-Set – Tuple - Dictionary.

Detailed Syllabus (Lab)

Detailed Syllabus(Lab)

DCA 105 - LAB I - Computer Basics and Office Automation

Lab Hours - 150 Hrs

Section 1 (25 Hrs)

1. Identify the various hardware components.
2. How to configure different peripheral devices.
3. Installation of operating system (Windows and Linux)
4. Familiarization of Desktop icons, Control panel, taskbar
5. How to create files, folder etc
6. User account management-create user, modify user, password setting, delete user etc
7. Familiarization basic Linux commands.

Section 2 (25 Hrs)

1. Identification of various network devices.
2. Practice to setup an IP address and DNS.
3. How to setup an internet connection with computer.
4. How to download a birth/death certificate from online?
5. How to add a certificate in Digi locker?
6. Perform a mobile recharge(using internet banking/UPI/Mobile wallet/Cards etc)

Section 3 (100 Hrs)

For Microsoft Word and Open Office writer

1. Create a two paragraph document and save the file, and perform cut,copy,paste,undo redo operations
2. Open a word file and perform text formatting operations (Bold, italic, underline,strike through, colour change ,background change etc)
3. Create a table with 5 column and 10 rows with fields and formatting(alignment, wrap text, border style etc) the table contents.
4. Create numbers and bullets in word document

5. Perform paragraph alignment, margin alignment, setup header and footer, page numbers in a document
6. Insert shapes, images, watermark etc in a word document
7. Create a two column document
8. Perform mail merge
9. Create and design an admission form and take a print out
10. Familiarise with unicode and asci Malayalam fonts
11. Familiarise inscript Malayalam typing
12. Perform typing in Malayalam various tasks like timetable, invitation, official and personal letters, etc preparation in word and writer
13. Prepare slides in malayalam

For Microsoft Excel and Open Office Calc

1. Creating simple Worksheet-Entering Text, numbers, Dates and Times ,etc
2. Perform column insertion, row insertion, column deletion, row deletion etc, comment insertion
3. Perform Formatting - Changing Character Style and character size, Changing Alignment, Changing row and Column Width, formatting decimal places, auto fill, formatting dates, Copying style and formats.
4. Insertion of header, footer, page numbers in worksheet.
5. Perform Copying entries between work books, Moving sheets between work books, Deleting sheets, cell references, quitting Excel.
6. Naming cells and Ranges
7. Perform simple calculations, Doing Arithmetic, Totaling column of values, Auto sum
8. Working of functions- Statistical functions, Mathematical functions, Text functions, Logical functions, Financial functions
9. Perform chart related works- Plotting charts, Sizing and moving charts, Updating charts, changing the chart type, Using chart auto format.
10. Perform sorting of data, , finding records, Adding and deleting Records, Filtering Records
11. How to create a password protected workbook?
12. Perform Mail merging using worksheet data.

For Microsoft PowerPoint and Open Office Impress

1. Creating a simple presentation, open an existing one, save and close a presentation
2. Perform basic functions like create a slide, insert, modify and delete a slide
3. Creating a Presentation with a Template. Creating a presentation from Scratch, deleting a slide, copying a slide, editing a slide
4. Perform changing font size and style, Changing the characteristics of Bullets, Changing indent, change slides background colour, copying slide, duplicate slide, Rearranging slides, Applying themes to slide layout
5. Inserting pictures, Shapes, Word Art, chart, header and Footer, Grouping objects in a slide
6. Applying transitions and animations in slides- Add and remove transition, Controlling Transitions between slides, Animating Different Parts of a slide
7. Insertion of audio and video in slides
8. Perform a slide show for a presentation

DCA 106-LAB II- Programming in C and Python & Web Application Development (HTML&CSS, JavaScript and PHP) **Lab Hours - 150 Hrs**

Section 1 – System Programming (70 Hrs)

Part A – Programming in C (50 Hrs)

1. Practice to write algorithm and draw flow chart for simple programs.
2. Programming on Basic Operations.
3. Programming Exercise on Conditional Statements: if, if else, switch.
4. Programming Exercise using loops : for, while, do-while and understand the concept of break and continue statements.
5. Programming Exercise using functions
6. Programming Exercise to understand Arrays.
7. Program based on basic string operations.

Part B – Python Programming (20 Hrs)

1. Programming exercise on Basics, Array, Operators.
2. Programming based on loops.

3. Programming exercise using Conditional Statements and Control Flow.
4. Programs based on strings.
5. Programming exercise on Data Structures - list, tuple, dictionary and set.

Section 2 - Web Application Development (80 Hrs)

Part A - HTML & CSS (40 Hrs)

1. Practicing basic HTML tags, font tags, paragraph styles, headings, lists
2. Tables in HTML, Frames in HTML, nested frames, Link and Anchor Tags
3. Creating marquee in HTML.
4. Programs to including graphics, video and sound in web pages
5. Programming exercise to create web pages.
6. Basic CSS programs.
7. Programs based on CSS styling background-text, fonts, links
8. CSS Programs using border, outline, margin
9. CSS programs using align, positioning and gradient.

Part B - JavaScript (20 Hrs)

1. Programs based on JavaScript operators
2. Programs based on conditional statements: if, if else, switch
3. Programs based on loops
4. Programs based on functions
5. Program to perform simple form validation

Part C - PHP (20 Hrs)

1. Programs based on basic data types.
2. Programs based on various operators.
3. Programming Exercise on Conditional Statements: if, if else, switch.
4. Programming exercise to create a simple web page using scripting.

Assignments for Theory And Questions for Lab

DCA - Paper 1(Theory)

DCA 101-INTRODUCTION TO INFORMATION SYSTEM

Module I - Computer Fundamentals

Assignment

1. History and Generations of Computers.
2. Explain in detail about various Input Output devices
3. Explain about different types of storage devices with capacity (based on purpose)
4. Specifications for new personal computer purchase
5. Show the computations needed for the conversion between various number systems

Module II - Basics of Information Technology

Assignment

1. Explain how information systems are related with modern life
2. Role of hardware and software in computing
3. Explain the role of information systems in education
4. Discovering the milestones of Information & Communication Technology history

Module III - Basics of Software

Assignment

1. Discuss the pros and cons of the three major operating systems.
2. What are the different versions of Windows OS
3. Updating operating system and web browser software
4. Explain about various DOS and Linux commands
5. Various methods for backing up of user data and disc management
6. Computer Virus and Green IT

Module IV - Basics of Information System

Assignment

1. Explain how is an information system is helpful for an organization
2. Explain the role of business information systems
3. Write the advantages of MIS
4. Explain an Enterprise System

Module V - Basics of Databases

Assignment

1. Write the importance of Data Processing
2. What are advantages of Data Management Systems
3. Explain various database management tools
4. Explain Relational Data Model Concept

DCA - Paper 2 (Theory)

DCA 102- NETWORKING AND INTERNET

Module I - Computer Networks

Assignment

1. Explain the various broadband technologies available for Internet access
2. Write the disadvantages of social media. What are the different ways to avoid the disadvantages of social media?
3. Compare GSM and CDMA standards
4. Explain the advantages of a smart card.

Module II – Introduction to Internet

Assignment

1. Explain any two types of web browsers
2. Briefly explain the concept of cookies, browsing history, parental control
3. List out the different types of email service providers and explain any two
4. Explain the importance of google class room in modern education

Module III – e-Governance and e-Commerce

Assignment

1. What are the major e-Commerce websites in India and explain?
2. Explain the idea of e-Office
3. What are the benefits of e-Governance to society?
4. What are the services provided by UMANG and Digi Locker?
5. What are the services provided by kerala.gov.in?
6. Write the importance of Citizen service portal

Module IV – Mobile Computing and Digital Payment System

Assignment

1. Explain in details about various outdated mobile operating systems
2. Explain the advantages and disadvantages of using mobile
3. Explain the concept of BHIM and Aadhaar Enabled Payment System (AEPS)
4. Explain about payment KIOSK
5. Explain the concept of ATM and CDM

Module V – Social Media & Cyber Security

Assignment

1. Explain the elements of cyber security
2. Explain the impact of social media in modern life
3. Explain the working and the advantages of Digital Signature
4. How to market a product using social media?
5. How to provide privacy and security features in your different social media accounts
6. Explain amendments in IT act.

DCA - Paper 3 (Theory)

DCA 103 - OFFICE AUTOMATION SYSTEMS

Module I - Word Processing and Indian Language Computing

Assignment

1. Malayalam / Hindi Paragraph typing
2. MS Word document preparation
3. Open office Writer document preparation
4. Preparing letters using Mail merge

Module II - Spreadsheets

Assignment

1. Preparing salary details of Employees
2. Different types of functions in MS Excel
3. Vlookup Applications with 2 files

Module III - Presentations

Assignment

1. What tools are available in PowerPoint to customize presentations?
2. Explain the steps to create animation and transition in PowerPoint.
3. How to change layout, header and footer in PowerPoint?

Module IV - Basics of Accounting & Billing Systems

Assignment

1. Explain basics of accounting
2. Explain Assets & Liabilities in accounting
3. Explain Features of Billing System
4. Explain billing using MS-Excel

Module V - Printing Technologies & Multimedia

Assignment

1. Explain various technologies used for printing
2. Explain image file formats
3. Explain Multimedia system architecture
4. Explain tools for integrating Audio with Video

DCA - Paper 4 (Theory)

DCA 104 – PRINCIPLES OF PROGRAMMING AND WEB TECHNOLOGIES

Module I - Basics of Programming

Assignment

1. Create Algorithms and flow charts.
 - a) Making instant coffee
 - b) Finding the biggest number among 3 numbers.
2. Programming language classifications and examples.
3. Object oriented programming concepts.
4. Microprocessor generations and types.
5. Differentiate Compiler and interpreter.

Module II - Introduction to C Programming

Assignment

1. Explain C Tokens.
2. Programs to illustrate the working of various Operators.
3. Explain various condition statements with examples
4. Explain looping statements with example programs.
5. Create a four-function calculator using switch-case.

Module III - C Programming: Arrays and Functions

Assignment

1. Single and multidimensional arrays creation, reading and printing, uses.
2. Explain any 5 string handling functions with example programs.
3. Create a four function calculator with functions add(),sub(), mul(), div()
4. Write a function to find the cube of a number.
5. Write a C program to find the factorial of a number using recursion.

Module IV - Web Programming: HTML, CSS and JavaScript

Assignment

1. Difference between HTML and HTML5
2. Explain any 5 HTML tags and their uses.
3. Create a Contact Us Form for a website using HTML
4. Create a Registration Page using HTML
5. Create a login page and perform form validation using JavaScript

Module V - Introduction to PHP and Python Programming

Assignment

1. Write a short note on various web servers.
2. Explain the features of the Python programming language.
3. Explain various looping and decision control statements in Python

DCA - Paper 5 (LAB)

DCA 105 – Computer Basics and Office Automation

Lab Sample Questions

Section 1

1. Perform the installation of printer in windows and Linux
2. Replace a damaged RAM in your computer
3. Insert a VGA card in correct port in the mother board
4. Create user named “CDIT” in windows 10 and set a password
5. Create a word document named “sample.docx” and save the file in “D:\test\
6. Create a folder “test” in linux and rename it as “test1” using command

Section 2

1. Connect your computer to internet
2. Download your birth certificate from website
3. Create Digi locker account add your driving license

Section 3

For Microsoft Word and Open Office writer

Question 1

Centre for Development of Imaging Technology, C-DIT, a Government of Kerala organization under Electronics and Information Technology, with corporate culture is concerned with customizing technology for the masses. Having its genesis in 1988 from a perception to use imaging technology as a tool to achieve progress, C-DIT has lived up to that dream.

- a) Type the paragraph above as it is using “Times New Romanst” , font size 12.
- b) Change the font type to “Book anitique” and size to 14
- c) Alignment to your paragraph to right margin.
- d) Save this file as C-DIT.docx in “My Documents” folder using save as option.
- e) Edit some part of your document and save again using save option

- f) Close the file.
- g) Open this file again using open option from office button.
- h) View the file you have created using print preview option.
- i) Take a print out of this document through print option.
- j) Open the above document
- k) Highlight the word "C-DIT"
- l) Search for word "kerala" using find option.
- m) Replace the word "C-DIT" with "C-DIT TED".
- n) Under line the word "C-DIT TED"

Question 2

Create document with following contents

"Microsoft office package includes MS word, MS PowerPoint, MS Excel and MS Access."

- a) View your document in portrait and landscape view using orientation option in page layout menu.
- b) Change the colour of the text using font colour option.
- c) Create a list as follows in table

1. MS Word	• MS Word	I. MS Word
2. MS Power point	• MS Power point	II. MS Power point
3. MS Excel	• MS Excel	III. MS Excel
4. MS Access	• MS Access	IV. MS Access

- d) Insert a picture below the above created table
- e) Insert a clip art in same document
- f) Insert different shapes
- g) Insert water mark "Sample" in this document
- h) Insert header and footer.

Question 3

Create a house warming invitation letter to your 5 friends, using mail merge

For Microsoft Excel and Open Office Calc.

1. Familiarization MS Excel/Open office calc interface
2. Create a workbook and enter text and numbers in worksheet
3. Create a workbook and enter data in the following format on the first sheet from the cell A1

	A	B	C	D	E
1	MONEY VALUES OF METALS				
2	Name	Symbol	Price/gm	Colour	
3	Copper	Cu	80.00	Golden Brown	
4	Gold	Au	4500.00	Yellow	
5	Silver	Ag	200.00	Grey	
6	Mercury	Hg	300.00	White	
7	Iron	Fe	100.00	Ash	
8					

- a) Adjust the column width to fit each data into the cell
- b) Change the format of price cells column with decimals & comma separator.
- c) Change the format of price cells with a \$ (dollar) currency style & also change \$ to ₹
- d) Find the average and sum of price field.
- e) Arrange the data in the ascending order of name and descending order of price
- f) Change the name of sheet to 'Metals'
- g) Change the font to 'Arial Black' with the size 12 for all the data entered
- h) Clear the format of contents in the sheet

Activate sheet2 & do the following.

- a) Fill the month names vertically from the cell A1
- b) Fill the day names horizontally from the cell B1
- c) Fill the range from C2 with numbers starting from 100 at regular intervals of 5 Upto C11.
- d) Save the Workbook as "Metals"

4. Simple Arithmetic-Add two numbers

5. Design a table on worksheet as follows

	A	B	C	D	E	F
1	No	Item	Unit Price	Quantity	Item stock	Total Value
2	1	Water	20	500	250	
3	2	Fanta	35	300	175	
4	3	Coca Cola	40	200	265	
5	4	Frooty	45	450	560	
6	5	Sprint	38	500	890	

- Define names on the column 3 and 4 correspondingly
 - Fill up the Total Value filed using names
 - Find out the sum of the total value of all items
 - Find out the average of unit price
 - Protect the worksheet with a password
 - Protect the workbook with a password
 - Save the file with name "list"
6. Extending an operation by clicking and dragging-Add 5 sets of two numbers
7. Absolute and Relative Referencing-Consider the problem of preparing a stationary order for the month of March. The item description, quantity and cost per item are available. The total cost per item is to be calculated and the final cost per item involves a sales tax of 2% over the total cost. The gross total and the net total are to be displayed.
8. Create an excel sheet to print the multiplication tables from 1 to 5 with each table ending at its 10th multiplication limit (i.e., $1 \times 10 = 10$.. $5 \times 10 = 50$).

9. Check the order details of a company shown below

	A	B	C	D	E	F	G	H
1	ORDER DETAILS							
2	ORDER ID	PRODUCT	UNIT PRICE	QUANTITY	DISCOUNT	REVENUE	TAX(2% FOR EACH ORDER)	NET INCOME
3	101	Brush	30	10	0%	?	?	?
4	101	Powder	50	25	0%	?	?	?
5	102	Umbrella	250	10	5%	?	?	?
6	102	Rain Coat	600	10	0%	?	?	?
7	102	Shoes	550	6	8%	?	?	?
8								

- Calculate the revenue and tax on the revenue for each product
- Calculate the net income of each product
- Calculate the total revenue of all products
- Calculate the total net income of all products

10. Check the following worksheet and find

	A	B	C	D	E	F	G
1	STUDENTS MARK DETAILS						
2	Roll No	Name	Maths	Physics	Chemistry	Sum	Average
3	1	Sunil	85	75	55		
4	2	Rahul	100	90	80		
5	3	Seetha	88	65	66		
6	4	Ajay	90	80	53		
7	5	Manu	95	90	78		

- Find the total and average of marks of 3 subjects of five students in a class in the board exam.
- Calculate the maximum mark, minimum mark and average for each subject using function

11. Create a worksheet containing the name and age of people. Count the number of people in each category as minor(<18), major(18-25), middle aged(26-40), above middle age (40-60) and senior citizen (61-100) using a single function for each scenario and print the count pertaining to the conditions.
12. Create a worksheet containing the consignments of different fruits from different suppliers. The fruit names in column A, supplier's names in column B, and quantity in column C. Find out how many quantities of each fruit is received.
13. See the below table

	A	B
1	Corporation Budget 2022	
2	Category	Amount in Lakh
3	Health	250000
4	Education	500000
5	Development	600000
6	Others	200000

- a) Make a Pie Chart
- b) Make a Line Chart
- c) Make a column Chart
14. Which settings are to be altered to type or enable Malayalam in OpenOffice.
15. Type any Malayalam text and format the same using Malayalam Inscript Keyboard for performing different character formatting commands from the toolbar, sidebar, and Character window.
16. Create a Mathrubhumi Calendar for any month. The calendar must be in the same format as seen and all the formatting must be applied.
17. Open Writer and prepare the following labels for
 - a) Schools to invite parents for annual function. The contents of the invitation must be in Malayalam. This must be in the form of a Letter.
 - b) Prepare a Meeting Agenda for School PTA Meeting and send it via email to the parents,

For Microsoft PowerPoint and Open Office Impress

1. Create a presentation with three slides
 - a) Apply any document theme
 - b) Add a title and insert slide contents
 - c) Save the presentation with a file name
2. Create a Presentation with a numbered Lists, add a title, format text and save.
3. Prepare a presentation to present the following data in two slides

Animation

- We can animate text, pictures, shapes and other objects.
- It means to add a special visual or sound effect to an object
- Zoom
- Bounce
- Fly In
- Float In
- Split

Type of Effects

1. Entrance
 2. Emphasis
 3. Exit
 4. Motion Path
- a) Add a new slide and insert a picture
 - b) Animate the slide contents with different effects.
 - c) Save presentation with a file name and make a slide show
 - d) Close the presentation
4. Create a blank presentation
 - a) Change slide layout
 - b) Add a title in word Art
 - c) Add slide contents

- d) Insert images in to the slide
 - e) Change background colour
 - f) Animate slide contents
 - g) Save and Run Presentation
5. Open any of your presentation
- a) Add proper Header and Footer
 - b) Add a new slide and insert some shapes
 - c) Include Date and Time to slides
 - d) Duplicate the first slide and change contents
 - e) Change the view to Slide sorted view
 - f) Check Spelling & Grammar
 - g) Change animations given to slide contents
 - h) Save and close Presentation
6. Create a presentation with at least 5 slides
- a) Apply any theme
 - b) Add text contents to the slides
 - c) Create a Watermark
 - d) Using shapes, create callouts, Banners and Stars
 - e) Add text to callouts
 - f) Apply Animations to slide contents
 - g) Reorder Animation
 - h) Check various presentation views
 - i) Save and Run presentation
7. Open previously saved presentation
- a) Change background colour and text colour
 - b) Check different transition effects and apply
 - c) Automate the presentation by specifying a waiting time of 2 seconds
 - d) Start slide show from beginning

8. Create a blank presentation with at least 3 slides
 - a) Add a banner title and contents to each slide
 - b) Insert an audio in to the third slide
 - c) Save and run presentation
9. Create a blank presentation with at least 2 slides
 - a) Add contents to the slides
 - b) Apply various animations to slide contents
 - c) Check various transition effects and apply
 - d) Add a video to your presentation
 - e) Save and run presentation
10. Create a custom slide show

DCA - Paper 6 (LAB)

DCA 106 -Lab II - Programming in C and Python and Web Application Development (HTML & CSS, JavaScript and PHP)

Lab Sample Questions

Section 1- System Programming (25 Hrs)

Part A - Programming in C

1. Write a C program to calculate simple interest.
2. Write a program in c to find the biggest number among three numbers
3. Write a program in c to calculate square of a number
4. Write a program in c to find even or odd numbers.
5. Write a c program to input marks from 5 subjects find the sum and average
6. write c programs to generate following patterns.

a) 1
 1 2
 1 2 3
 1 2 3 4

b) *
 * *
 * * *
 * * * *

7. Write a c program to print even numbers between 1 and 100.
8. Write a program in c to find the grade of student.
9. Create a function sum and perform addition operation.
10. Write a c program to print values in an array.

Part B – Python Programming

1. Write a program to print “helloworld”
2. Write a program to add two numbers
3. Write a program to finding biggest among two numbers
4. Write a program to find simple interest
5. Write a program to print even numbers between 1 to 100
6. Write a program to concatenate two strings
7. Write a program to accept a string from console and print it in reverse order
8. Write a program to print the given list after removing all duplicate values with original order reversed list=12,24,35,24,88,120,155,88,120,155,230,545,23,454,88
9. Write a program to add members in a vegetable set.
10. Write a program to Write a Python program to concatenate the following dictionaries with values
dic1={1:10, 2:20}
dic2={3:30, 4:40, 5:50}
dic3={6:60,7:70, 8:80, 9:90 } to create a new one.
11. Write a Python program to find the index of an item of a tuple.
12. Write a Python program to convert a tuple to a dictionary

Section 2 – Web Application Development

Part A – HTML&CSS

1. Write a program to display “Helloworld”
2. Create a webpage that prints the numbers 1 - 10 to the screen
3. Write 2 paragraph about yourself and setup heading “Who am I” using appropriate tags
4. Print a paragraph that is a description of a book, include the title of the book as well as its author. Names and titles should be underlined, adjectives should be italicized and bolded

5. Create a webpage and set its title to "This is my page"
6. Print two lists with different versions of Windows OS and Android. One list should be an ordered list, the other list should be an unordered list
7. Create a web page with link name "C-DIT" and reference goes to www.cdit.org
8. Display five different images. Skip two lines between each image. Each image should have a title.
9. Write an HTML program to display your SSLC marks in tabular form.
10. Create website that collects your personal information
11. Write a simple program to display your name
12. Write a program to display a paragraph at center
13. Write a program to display heading
14. Write a "hello world" program with background blue
15. Write program to display the word "Hello world" in different types borders
16. Write a program to set margin size 70px
17. Write a program to display "helloworld" in solid red border and green outline

Part B – JavaScript

1. Write a program to find the sum of three numbers
2. Write a program to find the product of two numbers using function.
3. Write a program to find the characters with in a string.
4. Write program to print numbers from 1 to 100 using for loop.
5. Write program to print first 100 counting numbers using while loop.
6. Write a program to show text "Javascript" when click on the button "clickme"
7. Write a program to display date and time when click on the button "time".

Part C – PHP

1. Installing PHP on Windows and Linux.
2. How to write PHP Scripts and Simple Hello World program?
3. Write a PHP program to display the today's date and current time
4. Write a PHP program to calculate sum of given number
5. Write a PHP program to read the employee detail using form
6. Develop a PHP program for email registration